

Supplementary Note 3. Influence of insulating substrate acidity on 3 mol% Gd-doped ceria (GDC3) nanowire arrays ionic properties.

In the case of GDC3, where ionic conduction is the dominant transport mechanism, which occurs through oxygen vacancy hopping, the conduction behavior differs from that of PCO_{NA}. Specifically, when a relatively basic MgO was used as a substrate, oxygen vacancy depletion was induced, leading to a total decrease in conductivity. On the other hand, the Al₂O₃ (relatively acidic) substrate promoted vacancy accumulation, resulting in a total increase in conductivity. In contrast to PCO, where bulk transport is composed of contributions from thermal carrier generation and their migration, in GDC3, the bulk conductivity is composed only of the migration energy of the oxygen vacancies (- 0.7 eV) as the carrier concentration is fixed by the aliovalent Gd³⁺ dopant, while for polycrystalline samples, activation energies > 1 eV are typically reported, associated with the additional grain boundary series resistance caused by the space charge barriers with positive potentials barriers^{11,13}.

The ionic properties of GDC3 nanowire arrays on two different substrates were characterized through Arrhenius plots in **Supplementary Fig. 10**. In this case, both samples were annealed at 400 °C, so the GB space charge potential driven by cation in-diffusion from substrate can be ignored. GDC3 nanowire arrays on the MgO substrate exhibited similar E_a (1.23 eV) comparable to that of polycrystalline GDC3 thin film. However, GDC3 nanowire arrays on the Al₂O₃ substrate exhibited a lower E_a of 1.01 eV. Moreover, GDC3 nanowire arrays on the Al₂O₃ showed higher in-plane conductivity than those on the MgO substrate.

These results are attributed to the conduction pathways determined by the acidity-mediated space charge potential at the heterointerface. In the case of MgO, a resistive heterointerface is formed, causing conduction to occur through the bulk (R_1). On the other hand, in the case of Al₂O₃, oxygen vacancy accumulation facilitates conduction through the substrate heterointerface (R_2), which is opposite to the tendency observed in PCO_{NA}. Furthermore, for Al₂O₃, the electric fields generated by the space charge at the heterointerface (due to a negative space charge potential) are opposite directions to the positively charged space charge at the GB, partially compensating each other and leading to a reduced E_a at the grain boundary heterointerface triple junction.

Supplementary Note 4. Comparison of polycrystalline and epitaxial PCO20 films to elucidate grain boundary in-diffusion effects

Polycrystalline and epitaxial PCO20 films were fabricated using PLD with a PCO20 target on Al₂O₃ substrate by modulating the temperature and deposition rate. For the polycrystalline PCO20 film, deposition was conducted at room temperature with a laser frequency of 10 Hz. In contrast, the epitaxial film was deposited at 700 °C with a reduced frequency of 2 Hz, while all other deposition conditions were maintained constant.

Evidence for the role of GB space charge potentials in impacting the effective conductivity of polycrystalline films can be obtained by comparing the AC impedance analysis of polycrystalline and epitaxial PCO20 films fabricated by PLD. We note that similarly grown polycrystalline and epitaxial films of PCO20 exhibited a trend (decrease in conductivity and activation energy) when moving from the epitaxial film to polycrystalline film, though the activation energies were generally lower than for the PCO20 bulk reference, likely due to lattice strain. In polycrystalline films, numerous GBs are present, allowing Al to diffuse into them during annealing. In contrast, in epitaxial films without GBs, such space charge modulation is prevented even under high-temperature annealing conditions (**Supplementary Fig. 12d**). As verified by XRD analysis (**Supplementary Fig. 12a and 12b**), the epitaxial films predominantly exhibit (111) and its family of planes, while polycrystalline films display a mix of planes, including (111) and (200). In addition, the shifted positions of peaks appearing in both polycrystalline and epitaxial films compared to PCO_{NA} imply tensile strain, which is reported to decrease the migration energy and improve diffusivity of oxygen vacancy hopping^{10,17}. As expected, the total conductivity of the epitaxial film is higher than that of the polycrystalline film due to increased blocking at the polycrystalline film's GBs. Furthermore, the Arrhenius plots reveal that the E_a of the polycrystalline film is lower than that of epitaxial films associated with the space charge regions surrounding the GBs, consistent with the PCO_{NA} results (**Supplementary Fig. 12c**).

Supplementary Table 1. Measurement of Pr chemical composition of PCO_{NA} fabricated using a PCO 20 at % target, through inductively coupled plasma mass spectrometer (ICP-MS). The reported loading amount of Pr in the PCO_{NA} represents the mean $\pm$ standard deviation ($n = 3$)

Sample	ICP-MS (at %)
PCO _{NA} (Fabricated using PCO 20 at % target)	19.15 $\pm$ 0.1